



Technology Trends Next 10–15 Years (By ~2030, 2035, 2040)

Over the next decade and a half, we expect rapid breakthroughs across many fields. In the **next 5 years (by ~2030)**, trends include the maturation of AI/ML and edge-cloud computing, pervasive connectivity (5G/IoT), and initial inroads in biotech and clean tech. By **10 years out (~2035)** we'll see AI-driven autonomy, quantum computing breakthroughs, immersive AR/VR, and advanced biotech. In **15 years (~2040)** a *hyperconnected*, AI-powered world will be normal: nearly every industry (healthcare, transport, agriculture, etc.) will leverage advanced AI, novel materials (like graphene), and biotech to transform life [dni.gov](#) [weforum.org](#) . Below we break out key trends in 5-year increments and highlight how governments and industry are prioritizing them – advice especially useful for an aspiring entrepreneur.

Next 0–5 Years (2025–2030) – AI/Cloud Everywhere

- **AI & Automation (Industry 4.0):** AI/ML will continue expanding into all business processes. McKinsey projects on the order of *50+ billion connected devices* by 2025 (mostly in industrial IoT), generating huge data for AI-driven automation [weforum.org](#) . Robots and smart sensors will handle routine tasks in factories, logistics and services. (For example, advanced robotics and 3D-printing are already producing massive data for optimization [weforum.org](#) .) However, note that U.S. survey data (2025) show AI adoption by large firms leveling off or even slightly declining [apolloacademy....](#) , implying companies are being more strategic about AI rollout. In practice, entrepreneurs should focus on *applied AI* – embedding ML into existing products and processes (the “AI as cost-of-business” notion) [pluralsight.com](#) – rather than chasing hype.



- **Connectivity & Cloud:** 5G rollouts and the coming 6G research will dramatically increase bandwidth and low-latency links. By 2025–2030, standards like 5G-Advanced/6G and satellite constellations (LEO) will enable “zero-latency” networks [pluralsight.com](#) . The EU’s digital strategy already aims for **75% of businesses using cloud-edge platforms by 2030** [digital-strategy...](#) . Most computing will become distributed: McKinsey/WEF estimate ~70% of firms will operate hybrid/multi-cloud architectures for flexibility [weforum.org](#) . Edge computing will process data on-device (target: 80% of data at the edge by 2025 [digital-strategy...](#)), so entrepreneurs should leverage cloud/edge services and distributed-infrastructure models. This also ties into **cybersecurity & trust:** as more data moves to the cloud and devices proliferate, “zero-trust” architectures and blockchain-based trust frameworks will be vital [weforum.org](#) [pluralsight.com](#) . In practice, expect strong demand for cloud architects, DevOps/site reliability skills, and security engineers.
- **Software & Development Tools:** Software engineering itself is changing. Trend reports predict that by 2030 *AI-powered dev tools* (e.g. code-generating assistants, low/no-code platforms, Infrastructure-as-Code) will become widespread [pluralsight.com](#) [weforum.org](#) . We’ll see “Software 2.0” where neural nets help write and optimize code. Developers should study ML-assisted programming (e.g. GitHub Copilot) and cloud-native architectures now to stay ahead.
- **Biotech & Health:** Advances in genomics, AI-driven drug discovery and synthetic biology will accelerate. AI+DNA sequencing breakthroughs are leading a **Bio Revolution** that touches agriculture, medicine and materials [weforum.org](#) [pluralsight.com](#) . Gene therapies and personalized medicine are on the cusp: e.g. within 5 years we expect new CRISPR-based treatments and mRNA platforms. Entrepreneurs can explore biotech startups or AI for drug design, as governments and big pharma boost R&D.
- **Climate & Clean Tech:** Rapid innovation in renewables, energy storage and climate tech is expected. Solar and wind will continue to get cheaper, and breakthroughs in batteries or carbon capture (e.g. via AI-driven materials design [weforum.org](#)) may happen. EVs and smart grids will scale (pointing to the “future of energy” trend). McKinsey notes clean-tech adoption will expand significantly, providing green energy to fuel the AI/data center boom [weforum.org](#) . Governments worldwide (EU Green Deal, US infrastructure bills, China’s clean energy plan) are heavily funding these areas, so entrepreneurs should monitor policy incentives in carbon/energy solutions.

- **Immersive AR/VR & “Metaverse”:** Significant VC and R&D dollars are flowing into augmented/mixed reality. By 2025–2030 expect advanced AR glasses and VR workspaces for remote collaboration. Pluralsight predicts “billions pumped into AR/VR startups” enabling remote work and training pluralsight.com. Meanwhile, Web3 concepts (blockchain, NFTs, digital identity) overlap with this: secure digital identities and decentralized networks (e.g. blockchain-based supply-chain or IoT networks) will grow weforum.org pluralsight.com. An entrepreneur with web/app skills can benefit by learning AR dev (Unity/Unreal) or blockchain protocols.
- **Embedded Hardware & IoT:** Count on a surge in smart sensors and specialized chips. “AI at the edge” (e.g. AI inference on device) will drive demand for AI accelerators, FPGAs, and neuromorphic chips. Governments (US CHIPS Act, China’s 5G IoT plans english.www.gov.cn) are investing heavily in semiconductor manufacturing and IoT device capabilities. Although core CPU scaling slows, expect growth in domain-specific hardware (AI chips, quantum bits, sensors). Hobbyist/entrepreneurs should consider combining skills in embedded C/C++, hardware design (FPGA/ASIC basics), and AI.

~5–10 Years (2030–2035) – Autonomy, Quantum and Beyond

- **Autonomous Systems:** Cars, drones and industrial robots will move from pilot projects to everyday use. Pluralsight labels the next phase “ACES vehicles” (Autonomous, Connected, Electric, Shared) as the new norm pluralsight.com. Expect widespread self-driving trucks and robot logistics, especially in developed markets. Robotics will also enter services (e.g. delivery drones, warehouse robots). As an entrepreneur, long-term transportation and robotics are huge opportunities, but heavily regulated; focusing on niche solutions (e.g. AI-powered fleet management, or robotics for elderly care) can pay off.
- **Quantum Computing:** Significant progress is expected by 2035. The roadmap in many labs (IBM, Google, EU Quantum Flagship) is to achieve *fault-tolerant quantum hardware* around mid-2030s. When it arrives, quantum AI algorithms could solve optimization, chemistry and encryption problems far beyond classical machines pluralsight.com. This will revolutionize drug discovery, materials science and cryptography. Entrepreneurs should start gaining quantum-aware skills (quantum programming, post-quantum cryptography), and follow government/industry partnerships in this area (e.g. US National Quantum Initiative, EU programs).

- **Ultra-Low-Latency Networks & 6G:** By 2030–2035, 6G standards are likely rolled out. This means truly ubiquitous connectivity for remote sensing/telemetry and IoT. Concepts like “zero-latency” (very close to 0ms) and ubiquitous satellite internet will enable new applications (e.g. remote surgery, autonomous vehicle platooning). The infrastructure buildout (dense small cells, advanced antennas) is underway – tech entrepreneurs can look into 6G PHY/MAC, advanced antennas, and integrating terrestrial/satellite nets.
- **Immersive Tech & Human Interfaces:** Beyond simple AR/VR, expect **brain-computer interfaces** and advanced wearables by 2035. DARPA and companies like Neuralink are actively developing BCIs pluralsight.com . Consumer AR glasses may become as common as smartphones. Meanwhile AI avatars and virtual assistants will become remarkably human-like (the Pluralsight vision of multimodal AI avatars pluralsight.com). Workplaces, education, and entertainment will blend with virtual worlds. Entrepreneurs should watch for platforms enabling AR/VR apps and health-tech interfaces, as well as AI/ML frameworks for multimodal interaction.
- **Advanced AI & Data:** Deep learning will be routine in business intelligence and R&D. By 2030 almost every software system will have AI under the hood (the “expected cost” of business pluralsight.com). We may see early Artificial General Intelligence (AGI) prototypes or “agentic AI” that can plan tasks end-to-end mckinsey.com . Data will be king: effective data pipelines, privacy-preserving ML, and AI governance tools will be in high demand. The recent US AI plan emphasizes **AI infrastructure** – from semiconductor fabs to national data sets whitehouse.gov . Smart entrepreneurs will develop skills in ML Ops, AI ethics, and domain-specific AI (e.g. AI for climate, AI for finance).
- **Biotech & Medicine:** Gene therapies, personalized medicine and bio-manufacturing will mature. By 2035 “hyper-personalized” treatments (tailored to your genome) may be common, driven by AI analysis of vast medical data weforum.org . Synthetic biology will create new materials (bio-plastics, lab-grown meat) and bio-sensors. Preventative medicine (AI diagnostics) will reduce healthcare costs. This is a massive field: startups in biotech, health AI, or agri-tech (GMOs, protein alternatives) will find venture and government grants (e.g. “Bioeconomy” initiatives).

- **Sustainable Technology & New Materials:** Sustainable tech will be essential infrastructure by 2030s. Expect breakthroughs like large-scale carbon removal, next-gen nuclear, or ultra-efficient batteries, spurred by AI-driven materials discovery. Novel materials (graphene, 2D materials, etc.) will enter high-volume manufacturing [weforum.org](https://www.weforum.org). For example, graphene-based chips or ultra-strong composites could become common. Entrepreneurs should monitor IP around new materials and seek partnerships in “cleantech startups” as energy and materials production is revolutionized.

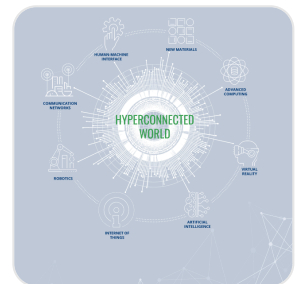
~10–15 Years (2035–2040) – Hyperconnected, AI-Driven World

Hyperconnected AI Society: By 2040 we are likely to live in a world where AI and connectivity are as ubiquitous as electricity.

DNI forecasts a “*hyperconnected world*” by 2040 – orders of magnitude more devices and data linking all aspects of life dni.gov.

AI will permeate daily routines (transport, education, agriculture, etc.), dramatically improving productivity and quality of life dni.gov.

Robotics and automation will have displaced many manual jobs, but created new fields in AI management and human–machine teaming.



- **Artificial General Intelligence (AGI):** It’s uncertain if true AGI will arrive by 2040, but narrow AI systems will be extremely advanced. AI assistants may have human-like abilities in language, reasoning, and creativity. Alongside this, society will wrestle with AI governance and ethics at scale (e.g. privacy, bias). Entrepreneurs should note that even by 2040, there will be a *premium* on integrating human insight with AI – e.g. neurotech, explainable AI tools, or AI audit services will be big areas.
- **Biotech and Longevity:** Aging research may yield treatments that significantly extend healthy lifespan. By 2040 biotech could have largely solved many cancers, genetic diseases, and created lab-grown organs dni.gov. Personalized nutrition and microbiome engineering could be mainstream. Think of health as a technology stack: IoT health monitors, AI diagnostics, and on-demand gene editing. This means entrepreneurs should follow biotech trends now – drug platform startups, health-focused AI products, and even wellness tech (mind-computer interfaces, biotech wearables).

- **New Industries – Space and Beyond:** Space exploration and related industries will likely expand. By 2040 China and private companies (e.g. SpaceX) aim for lunar bases and asteroid mining [dni.gov](#). Satellite internet and tourism will create billion-dollar markets. Robotics and AI will be key in space (autonomous rovers, AI navigation). Similarly, “smart cities” with IoT sensors for traffic, energy and climate control will mature worldwide. Entrepreneurs can look to space-tech (ground segment services, cubesat platforms) and smart-city startups (urban data analytics, infrastructure AI).
- **Energy and Climate Reality:** We expect a mostly decarbonized grid by 2040, with massive renewable fleets and smart grids (AI-optimized). Geoengineering or fusion (if breakthroughs occur) may be real options by then. Those trends are driven by urgent climate needs and government action. For entrepreneurs, this means energy storage, carbon capture, sustainable agriculture, and climate risk analytics will be key industries.
- **Materials & Manufacturing:** Advanced manufacturing (3D-printing of complex structures, self-repairing materials) will be widespread [dni.gov](#). Cheap, strong materials (e.g. graphene electronics) will enable devices we can’t yet imagine. Factories may be run by AI with minimal human oversight. Hardware entrepreneurs should therefore emphasize flexibility and learning new manufacturing tech. For example, skills in additive manufacturing, nano-fabrication, or bio-fabrication will be in demand.

Global Focus and Entrepreneurial Outlook

- **Government Priorities:** The major economies are strategically investing in these technologies. China’s 2026–2030 plan, for instance, emphasizes “breakthroughs in core tech” – specifically AI (new models, high-end chips), semiconductors, quantum, and biotech [english.www.go...](#) [english.www.go...](#). The U.S. (via its National AI initiative and CHIPS+ Science Acts) is similarly funding AI research, semiconductor fabs and computing infrastructure [whitehouse.gov](#). The EU is investing in cloud/edge infrastructure and green tech (e.g. aiming to triple data-center capacity in 5–7 years [digital-strategy....](#)). As an entrepreneur, align your skills and ventures with these priorities. For example, startups in AI, renewable energy, or health tech may qualify for grants or subsidies.
- **Big Tech and R&D:** Large tech companies continue pouring R&D dollars into emerging fields. Google/DeepMind, Microsoft/AI, Amazon/space (Kuiper) and others are labs-to-market engines. Monitor their announcements – they often set de facto direction (e.g. Google’s emphasis on on-device ML, Tesla on self-driving, Apple on AR/health). You can ride these waves by providing niche solutions or services to big players (e.g. AI model improvement tools, cybersecurity for AI).

- **Startup and Investment Trends:** Even if some hype cycles (like AI) mature, venture capital remains active in frontier areas. For example, despite a slight slowdown in enterprise AI adoption, VC funding is strong in **digital-trust (zero-trust, blockchain), quantum, agtech/foodtech, and clean tech**. Endeavor and others report record VC deals in biotech and space in late 2020s. Keep an eye on accelerator programs (e.g. XPRIZE, EU IPCEIs, DARPA challenges) – many emerging tech fields have public prize programs.
- **Global Markets:** “Where money flows” depends on regional strengths. In the **U.S./Canada**, fintech, biotech, and advanced AI (including defense tech) dominate. **China** is leader in large-scale deployments of 5G, smart cities, and e-commerce AI, so opportunities there include consumer AI applications and manufacturing tech. **Europe** stresses data privacy and green tech – think startups in sustainable manufacturing, energy, and EU-compliant cloud services. **India and SE Asia** are hot for software outsourcing, e-commerce, and frugal innovations (cheap sensors, Agri-tech). **Middle East/Africa** are investing in fintech/mobile-payments and renewable energy (e.g. solar for development). Be ready to adapt your product to local regulations (EU’s Digital Markets Act, China’s cyber rules, etc.) to seize global demand.
- **Personal Strategy:** Given these trends, here are some concrete tips:
 1. **Skill Up:** Deepen your expertise in AI/ML and cloud platforms (e.g. get proficient with TensorFlow/PyTorch, AWS/Azure/Google Cloud). Also learn data engineering, cybersecurity principles, and gain some hardware knowledge (e.g. Raspberry Pi/Arduino for IoT, FPGA basics).
 2. **Education & Networks:** Consider advanced courses or degrees in high-demand fields (AI safety, quantum computing, biotech). Participate in hackathons, conferences and tech meetups globally.
 3. **Entrepreneurial Angle:** Identify niches where demand outpaces supply. For example, if AI hype is waning, focus on underserved areas like **AI-driven sustainability solutions** or **secure AI tools**. Combine your interests: an “AI + Embedded” startup (e.g. smart IoT sensor with on-device AI) or “Web3 + Cloud” (decentralized cloud services) could stand out.
 4. **Leverage Funding:** Look for government grants/partnerships. The EU, US, China etc. fund strategic industries heavily. For instance, the U.S. funds startups via SBIR in AI/health, the EU has “Important Projects of Common European Interest” for cloud/edge and green tech [digital-strategy...](#), and China offers R&D tax breaks for high-tech firms [english.www.go...](#).

- 5. Stay Agile:** Tech evolves unexpectedly. Keep learning and be prepared to pivot. For example, even if AI becomes commodity, the *combination* of AI with other breakthroughs (like AI-driven genomics or AI-managed IoT networks) will create new markets.

In summary: Expect a rapid mix of **AI/ML, cloud/edge, connectivity, biotech, advanced materials, and clean energy** to drive the next 10–15 years. By mapping these trends onto your skills and the global market landscape, you can position yourself for success. Continually track tech reports (like McKinsey, WEF, DNI) and industry news, and use them to iterate on your strategy. With sustained learning and a keen eye on where governments and big tech invest, an ambitious entrepreneur can ride these waves to build transformative businesses.

Sources: Authoritative trend reports and government strategies (McKinsey/WEF weforum.org, weforum.org, Pluralsight/WEF pluralsight.com, pluralsight.com, DNI Global Trends dni.gov, dni.gov, EU digital strategy [digital-strategy...](https://digital-strategy.europa.eu), [digital-strategy...](https://digital-strategy.europa.eu), US AI plan whitehouse.gov, Chinese 5-year plan [english.www.go...](https://english.www.gov.cn), [english.www.go...](https://english.www.gov.cn), etc.) were used to identify and timeline these tech trends. Each cited source underpins key predictions above.

